

are stored! If both of them run separately, they'll have their own (virtual) disks and won't work, right? Yes, they won't work but only if you forget that the disks that those virtual machines will use are also virtual. With good virtualization software, you can mount remote disks. So the second email server might just be able to use the first email server's disks as well! There can be different types of configurations by which this can be done, so much so that they're beyond the scope of this little book. It's just to make the point that the resources used by virtual machines are also virtual and they can be used in various ways and one of those ways is to share the resources.

► **Recovery:** Let's say an administrator just made a mistake and accidentally made some changes to the machine configuration and now he and the other users are unable to find out what the change was! This can be a disaster without virtualization. Depending on what went wrong, what was destroyed and how important it was, virtual machines can help ease your troubles. Here are some examples:

- If you have a 'misconfiguration' issue and have cloned virtual machines then you can simply copy back the configuration!
- If it was data and you didn't have backups, but do keep regular snap shots, the virtual machine will reduce the heat you're going to take!
- If you have backed up or cloned disks, you don't really need to worry, do you?
- If the data which the virtual machine was working on was on a separate virtual disk than on which the OS of the virtual machine was installed, you can simply create a new one! (well, there are other complications but they're usually fewer than when you were not using virtual machines).

► **Fault-tolerance:** There are times when machines fail. They're meant to fail at some point in time! Virtualization can even help in such instances. Since all the hardware in use by the virtual machine is also virtual and can have multiple copies and at remote places, one server failing might not just bring down everything if you had other copies ready and running!

If you see now, it would be so very clear why virtualization is really good for servers. Because of these benefits, virtualization is slowly becoming the need for servers rather than a feature. One of the biggest things that virtualization brings to the world is cloud computing. While cloud computing doesn't really depend on virtualization, it does help in numerous ways.